

Ruben Taieb

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Education

- INRIA de Lyon and Le Havre University** France
Mathematics 2024 - 2027
- PhD in Mathematics, PDEs and dynamical systems applied to biology and population dynamics
- École Normale Supérieure de Lyon** France
Mathematics 2020 - 2024
- Master's degree in Mathematics (probability and statistics), minor in Theoretical Physics
- Lycée Masséna** France
Mathematics, Physics and Computer Science 2018 - 2020
- Preparatory classes for French grandes écoles

Educational background

Mathematics

Statistics, Machine Learning, Stochastic processes and calculus, functional analysis, PDEs

- Parametric and non-parametric statistics
- (Non)linear regression, Stochastic Modelling, AI theory, neural networks
- Mixed effects models
- Stochastic calculus
- Stochastic processes, Markov chains, brownian motion, random matrices
- Differential geometry, complex analysis/geometry, linear algebra, field/group theory
- Functional analysis and PDE theory
- Elliptic PDEs, transport PDEs, Hamilton-Jacobi PDEs

Physics

- Quantum mechanics
- Analytical mechanics, Hamiltonian, Lagrangian
- Statistical physics, thermodynamics
- Dynamical systems and chaos theory
- Solid state theory

Biology

- Immunology, Bacteriology, Virology
- Neurology and neurodevelopment

Experience

- INRIA, Team COMPO in Marseille, France** | 6 week research internship Jun – Jul 2021
- Mixed effects models for oncological/pharmaceutical data analysis with Sebastien Benzekry
 - Review of different tumour growth models
- INRIA, Ecole Polytechnique of Paris, France** | 8 week research internship May – Jul 2022
- Algorithms and methods for covariate model selection in clinical data analysis with Marc Lavielle
 - Study of SAEM and SAMBA algorithms for mixed effects model selection
- Potsdam University, Germany** | 16 weeks research internship April – Jul 2022
- Development of an algorithm for SDE parameters estimation in mixed-effects modelling, through SAEM

- Model misspecification diagnosis using SDEs in a mixed effects setting, for pharmacological purposes, with Niklas Hartung and Wilhelm Huisinga

INRIA and Roche/Genentech, France | *6 months research internship* October – March 2024

- Mixed effects - machine learning - survival analysis modeling of outcome in oncology for clinical trial optimization with Sébastien Benzekry
- Development of a mathematical, population model for tumour growth coupled with biomarker dynamics (albumin, LDH, neutrophil count) applied to survival analysis
- Presented a talk for PAGE conference

ENS de Lyon, France | *4 months research internship* April – July 2024

- Stochastic Galton Watson modelling of mutations in cell population under treatment, with Hélène Leman

INRIA Lyon and Le Havre University | *3 year PhD* 2024 - 2027

- Age-Structured, multi-strain, multiscale epidemiological models with reinfection with Mostafa Adimy and Arnaud Ducrot
- Differential equations coupled with non-local difference equations - Modelling of localized phenomena in population dynamics
- Modelling population dynamics with quiescent phases
- Study of an age-structured multi-strain nested immuno-epidemiological model for dengue reinfection

Programming: Python, R, Git

Languages: French, English, Portuguese (Fluent), German (intermediate)